

FARAG MOUSA ELABD

AI Engineer | Machine Learning & Deep Learning

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SUMMARY

AI Engineer deeply passionate about Computer Vision and Deep Learning. I enjoy building end-to-end solutions — from raw data to deployed models — that solve real problems. My goal is simple: keep building, keep learning, and use AI to make a real difference.

EDUCATION

Bachelor of Artificial Intelligence | Menoufia University — Dept. of Machine Intelligence July 2025

- Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Neural Networks, Data Structures & Algorithms, Mathematics for AI

TECHNICAL SKILLS

Programming Languages	Python, C++, SQL
AI/ML Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face, Transformers
Deep Learning	CNNs, RNNs, LSTM, GRU, GANs, Transformers, YOLO (v5/v8), ResNet, VGG, Transfer Learning, Data Augmentation, Batch Normalization
Computer Vision	Object Detection, Image Classification, Segmentation, Pose Estimation, Face Recognition, Feature Extraction, Edge Detection
Machine Learning	Supervised (Regression, Classification, Decision Trees, Random Forest, SVM, XGBoost), Unsupervised (K-Means, PCA, t-SNE), Hyperparameter Tuning, Cross-Validation
Data Science	NumPy, Pandas, Matplotlib, Seaborn, NLTK, spaCy, EDA, Statistical Analysis
Databases	SQLite, MySQL, Database Design
Tools	Git/GitHub, Jupyter, Google Colab, VS Code, PyCharm, Linux/Unix, Arduino, Raspberry Pi
Cloud & Deployment	Azure, Flask, FastAPI

SOFT SKILLS

Problem Solving	Communication	Leadership	Teamwork	Strategic Planning	Adaptability	Time Management
Emotional Intelligence						

PROJECTS

Fire & Smoke Detection System

Python · YOLOv8 · Computer Vision · Deep Learning

- Developed a real-time fire and smoke detection system using YOLOv8, merging multiple datasets and applying preprocessing to improve model generalisation.
- Built a complete pipeline: data preparation → training → real-time inference, designed for early hazard detection and monitoring applications.

Wind Turbine Temperature Prediction

Python · XGBoost · Time-Series Analysis — IEEE Competition

- Built a predictive model for rotor bearing temperature on real-world industrial data with extensive time-based feature extraction and preprocessing.
- Applied ML techniques to model complex environmental-mechanical relationships for predictive maintenance applications.

Irrigation Need Prediction System

Python · Machine Learning · Deep Learning · Feature Engineering

- Developed an AI system using agricultural and environmental data; applied advanced feature engineering to handle complex patterns and class imbalance.
- Built and evaluated multiple predictive models to support smarter water management, improving efficiency and reducing resource waste.

Gait Analysis System

Python · Computer Vision · Pose Estimation · Deep Learning

- Built a human gait analysis system tracking body keypoints across video frames using pose estimation, extracting motion features for posture and behaviour study.
- Applicable to healthcare, security, and performance analysis domains.

EXPERIENCE

AI & Robotics Instructor | Techno Future

Aug 2025 – Jan 2026

Damanhour, El-Behira

- Taught programming, robotics, and AI fundamentals to 50+ students across 3 courses through hands-on, project-based learning; achieved **90%+ student satisfaction**.
- Developed comprehensive curriculum simplifying complex technical concepts into accessible, practical sessions.

Training & Certifications

AI & Data Science Diploma — Digilions	End June 2026
Microsoft Machine Learning — DEPI / EYouth	Winter 2025
Artificial Intelligence Training (90 hrs) + Odoo Course — ITI	Summer 2024
C++, OOP, Data Structures — Route Center	Summer 2023
AI Technical Training (90 hrs) + Soft Skills (30 hrs) — NTI	Summer 2022

Volunteering & Leadership

PR & Fundraising Director

2023 – 2025

NASA Space Apps Shebin El-Kom

- Managed sponsorships, partnerships, and speaker coordination for 200+ participant hackathon events, securing \$15K+ in funding.
- Coordinated with 10+ judges and industry speakers through strategic outreach and relationship building, enhancing event credibility and participant learning.

Partnership & Sponsorship PR

2023 – 2025

GDSC MSB

- Organized collaborations with 8+ tech companies to support community initiatives, providing resources and opportunities for 300+ student members.
- Cultivated partnerships resulting in workshop sponsorships, mentorship programs, and job placement opportunities.

Partnership & Sponsorship Member

AIEC

- Contributed to partnership management, sponsor relations, and event coordination for community tech initiatives.